The FREUID Challenge 2026

Technical Report — junesdata

Necula George-Andrei
Independent
tinkode6@gmail.com

Report date: July 15, 2026

Abstract

We address binary fraud detection on identity-document images. Our key observation is that fraud signal in FREUID is dominantly *local*: manipulations live at the level of pixels and small regions, while document-level semantics occur in both classes. Consequently we avoid full-image downscaling entirely and train binary patch classifiers (EfficientNetV2-S and EfficientNetV2-M, ImageNet-21k pre-trained) on random 512×512 crops taken at native resolution, with a recapture-simulation augmentation (resampling, moiré-like patterns, sensor noise, JPEG recompression) targeting the capture-pipeline shift emphasized by the test set. At inference each image is covered by an overlapping grid of 512×512 crops and scored by the mean of its top-2 crop logits; the final score is the mean of per-model ranks, a scale-free combination well matched to the ranking-based FREUID Score (lower is better). Training uses only the official FREUID training set plus iterative pseudo-labeling on the official public test images. Full-image baselines scored 0.20–0.25 on the public leaderboard; the patch pipeline with pseudo-labeling and the two-model rank ensemble reached **0.00226**. Code (MIT), frozen weights, and a no-network Docker container reproducing the selected submissions are public.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026)
Kaggle team: junesdata
Code: https://github.com/4kaws/freuid-challenge-2026 (commit [FINAL-SHA])

1 Introduction

The FREUID Challenge [1] asks for a calibrated fraud score for identity-document images under a realistic threat model: physical manipulations, GenAI-driven edits, and print-and-capture forgeries that suppress fragile digital artifacts. The evaluation emphasizes generalization: the private test set contains document types absent from both training data and public test data, and places stronger weight on recaptured (non-synthetic) examples.

Early experiments showed that models trained on downscaled full images learn shortcuts that do not survive the distribution shift between the training generation batch and the test batch (public-leaderboard FREUID $\approx$ 0.20–0.25 despite saturated in-distribution validation). Manual inspection indicated that semantic anomalies (implausible fields, mismatched attributes) occur in *both* classes, i.e. they are generator noise rather than fraud signal, while true fraud is local pixel-level manipulation. This motivated a native-resolution patch approach.

2 Method

2.1 Overview

The pipeline is: native-resolution 512×512 patch classifier(s) $\rightarrow$ overlapping grid-crop inference with top-k logit pooling per image $\rightarrow$ cross-model rank averaging $\rightarrow$ submission scores in $[0, 1]$.

2.2 Model architecture

Two timm [2] backbones fine-tuned as binary classifiers (`num_classes=1`):

- `tf_efficientnetv2_s.in21k_ft_in1k` (≈21M parameters);
- `tf_efficientnetv2_m.in21k_ft_in1k` (≈54M parameters).

Inputs are 512×512 RGB crops normalized with ImageNet statistics. No resizing is applied anywhere: crops are taken from images at their original resolution (images smaller than 512 px on a side are zero-padded).

2.3 Training objective and procedure

Binary cross-entropy with logits; AdamW (lr 3×10^{-4} for V2-S with batch 48, 2×10^{-4} for V2-M with batch 24; weight decay 10^{-2}); OneCycle schedule; 4 epochs; mixed precision. One random 512×512 crop per image per epoch with horizontal/vertical flips. With probability 0.3 a *recapture simulation* is applied to the crop: random down/up-resampling (0.45–0.9×, mixed interpolation), moiré-like sinusoidal luminance patterns, Gaussian sensor noise, gamma/color drift, and JPEG recompression (quality 45–92). This augmentation targets the print-and-capture emphasis of the test distribution.

Iterative pseudo-labeling. The public test images (unlabeled competition data) are incorporated by self-training: after each round, images with fraud probability > 0.98 are pseudo-labeled fraud, and the 1,150 lowest-ranked images are pseudo-labeled genuine (rank-based rather than threshold-based, because confident genuine calibration collapses after the first round). Five rounds were run; each round retrains from ImageNet initialization on train + current pseudo-labels. Public leaderboard progression across rounds: $0.0587 \rightarrow 0.0350 \rightarrow 0.0224 \rightarrow 0.0038 \rightarrow 0.0035$ (with inference improvements applied along the way, see Section 4).

2.4 Score calibration

The FREUID Score is computed from the DET curve and an operating point (APCER@1%BPCER), both invariant to strictly monotone transforms of the scores. We therefore use rank-normalized scores: for each model, per-image aggregated logits are converted to ranks in $[0, 1]$ over the full test set; the submission score is the (weighted) mean of per-model ranks, clipped to $[10^{-6}, 1 - 10^{-6}]$.

3 Data

3.1 FREUID training set

All 69,352 training images across the five document types were used. For model selection during development we used leave-one-type-out folds (train on four types, validate on the held-out type), which correlates with the public leaderboard far better than random splits (full-image models: held-out-type FREUID ≈ 0.07–0.22 vs. in-distribution ≈ 0).

3.2 External data and pre-trained models

No external data beyond FREUID and publicly licensed pre-trained weights. Pre-trained backbones are the public timm ImageNet-21k/1k checkpoints (Apache-2.0) [2].

3.3 Validation strategy

Development signals: (i) leave-one-type-out validation on the training types; (ii) the public leaderboard, used as the primary signal for the pseudo-labeling rounds (the public test set is explicitly designated as a development signal by the organizers). Private test labels were never available; private test images were not used for any training or tuning — all model weights were finalized and archived (timestamped Kaggle dataset `junesdata/freuid-ckpts`) *before* the private test release on July 13, 2026, 07:02 UTC.

4 Inference

Each test image is covered by an overlapping $R \times C$ grid of 512×512 crops at native resolution (grid offsets evenly spaced; images smaller than the crop are zero-padded). Per model, the image score is the mean of the top-2 crop logits — max-style pooling is essential because fraud is local (mean pooling degrades the public score by $\sim 3\times$). Grid density improved the public score up to $7\times9 = 63$ crops, with saturation beyond. The submitted configurations balance accuracy against the 6-hour A100 verification budget.

Two final picks are produced from one Docker image and one frozen weight set, differing only in a documented environment flag (inference orchestration only):

- **Pick 1** `VARIANT=ens2`: rank-mean of EfficientNetV2-S (round 5) and EfficientNetV2-M (round 5), grid `[FINAL-GRID-1]`.
- **Pick 2** `VARIANT=hedge`: adds the round-1 pseudo-label model and the no-pseudo-label base model at half weight, grid `[FINAL-GRID-2]` — an out-of-distribution hedge that reduces specialization to public-test fraud patterns.

Commands and output checksums: see Table 1 and the repository README.

Table 1: Final picks: submission ↔ command ↔ checksum.

Pick	Command	`sha256(submission.csv)`
1 (`ens2`)	`docker run -network none ... -e VARIANT=ens2`	`[CHECKSUM-1]`
2 (`hedge`)	`docker run -network none ... -e VARIANT=hedge`	`[CHECKSUM-2]`

5 Results

Table 2: Public-leaderboard progression (FREUID Score; lower is better).

Configuration	Public FREUID ↓	Notes
Full-image EfficientNetV2-S @384/@768	0.250 / 0.202	shortcut learning
Patch 512, 2×3 grid, top-2	0.0737	
Patch 512, 3×4 grid, top-2	0.0587	
+ pseudo-labels (round 1)	0.0350	
+ denser grid 5×6	0.0276	
+ pseudo-labels (round 3)	0.0224	
+ round 4, 7×9 grid	0.0038	
+ round 5, 7×9 grid	0.0035	
+ two-model rank ensemble (V2-S + V2-M)	**0.00226**	
Final Pick 1 (`ens2`)	[TBD]	Selected final submission
Final Pick 2 (`hedge`)	[TBD]	OOD-robust variant

6 Reproducibility

- **Repository:** https://github.com/4kaws/freuid-challenge-2026, commit [FINAL-SHA] (MIT license).
- **Docker:** `docker build -f docker/Dockerfile -t freuid-solution .`; weights are inside the repository (Git LFS) and are copied into the image — no runtime downloads.
- **Dependencies:** PyTorch 2.3.1 + CUDA 12.1 base image; timm 1.0.11; OpenCV (headless); pinned in `docker/requirements.txt`.
- **Runtime:** single A100 40GB; measured [RUNTIME] for the full 142,818-image test set at the submitted configuration (within the 6-hour verification budget). CPU fallback exists but is not practical at scale.
- **Randomness:** inference is deterministic up to GPU non-associativity; no test-time augmentation with random components is used.
- **Network:** the container runs with `-network none`; verified locally before submission.
- **Code freeze:** all model weights used in the final picks were archived in the version history of the Kaggle dataset `junesdata/freuid-ckpts` before the private test release (last weight version: July 13, 2026, 04:56 UTC).

7 Team and contributions

Necula George-Andrei — modeling, data analysis, engineering, and writing.

Acknowledgments

We thank the organizers and the Microblink Fraud Lab for a well-designed competition.

References

[1] Ivan Relić, Vincenzo D'Elia, Stefano Bortoli, Radu Tudoran, Hristina Nedyalkova, Mihaela Bošnjak, Stanislav Pavlić, Tin Mavračić, Darin Dašić, Marin Kačan, Jerko Šegvić, Paolo Čerić, Filip Šoprun, Matej Miočić, Marin Oršić, Lorenzo Vaiani, and Paolo Garza. The FREUID challenge 2026: 1st competition on identity documents fraud detection. https://freuid2026.microblink.com/, 2026. IJCAI-ECAI 2026 competition.

[2] Ross Wightman. Pytorch image models. https://github.com/huggingface/pytorch-image-models, 2019. Apache-2.0 license.

A Hyperparameters

Hyperparameter	Value
Crop size	512×512 (native resolution, no resizing)
Batch size	48 (V2-S) / 24 (V2-M)
Learning rate	3×10^{-4} (V2-S) / 2×10^{-4} (V2-M), OneCycle
Weight decay	10^{-2}
Epochs	4 per round
Recapture augmentation prob.	0.3
Pseudo-label thresholds	fraud $p > 0.98$; genuine = 1150 lowest-ranked
Inference pooling	mean of top-2 crop logits
Ensemble	rank mean over models
Seed	42 (training); inference deterministic

B Negative results

For completeness: per-type rank normalization of a full-image model (0.83), mean pooling over crops (0.21), ensembling leave-one-type-out fold models into the test ensemble (dilutes: 0.075 vs. 0.035 baseline), agreement-gated pseudo-label round 2 (0.043), a 6th blind pseudo-label round without leaderboard feedback (0.0055 vs. 0.0035), and denser grids beyond 7×9 (flat: 0.0036 at 9×12).